Proactive to Reactive: AI-Powered Data Security Investigations (DSI) with Microsoft Purview

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Introduction to AI-Powered Data Security Investigations (DSI)

Why are data security investigations hard?

- Log centric and keyword driven
- Time-consuming and incomplete
- Limited visibility into what the data contains
- Data context and sensitivity is often overlooked

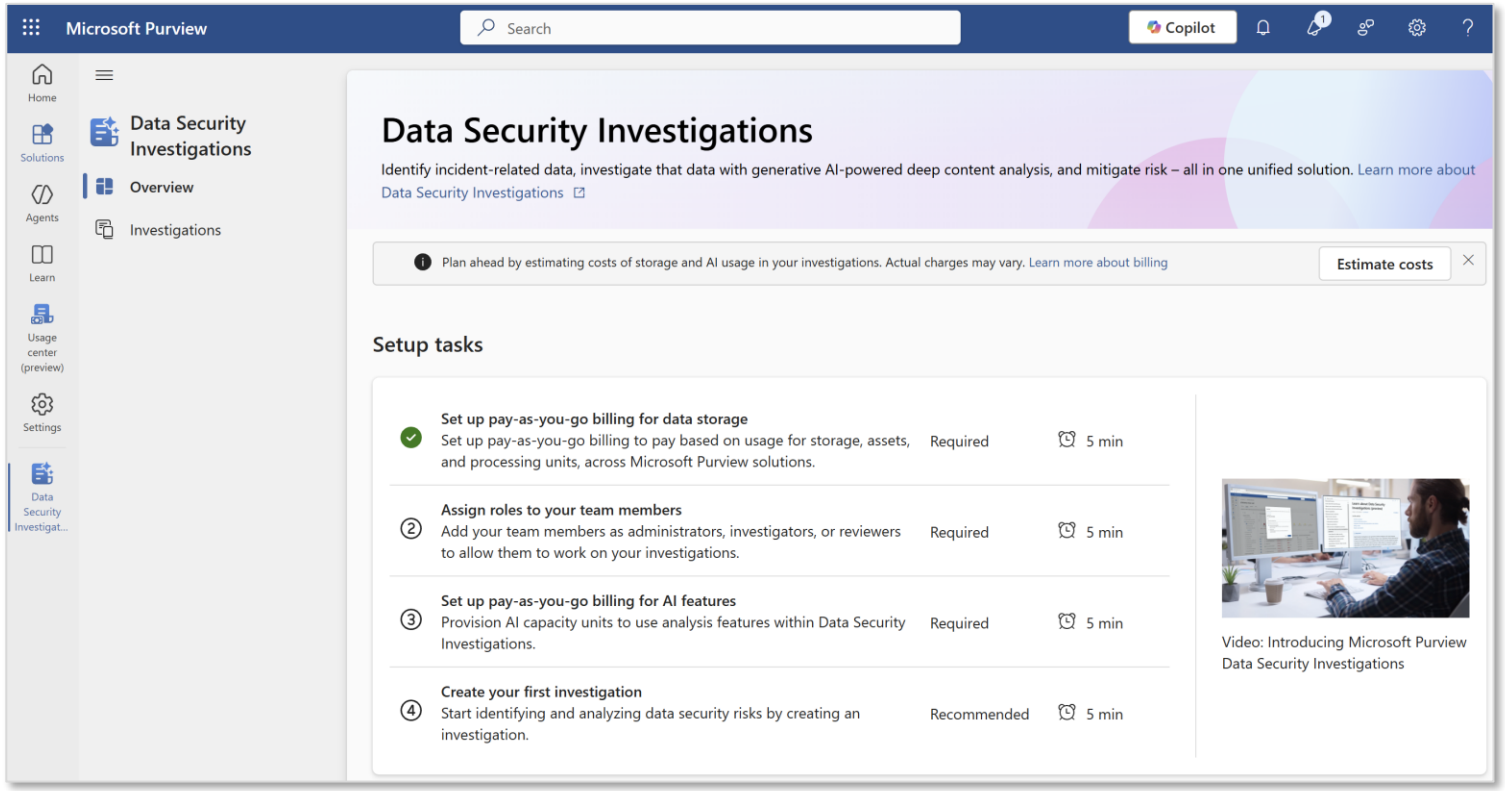
Missing Piece: Understanding the Data

- Understanding the types of data impacted can change the response
- Content and sensitivity determine business risk
- Without data understanding, impact is underestimated or overestimated

From Reactive to Proactive Security

- Reactive: investigate after a breach
- Proactive: identify and mitigate risk before an incident
- Modern tools must support both models

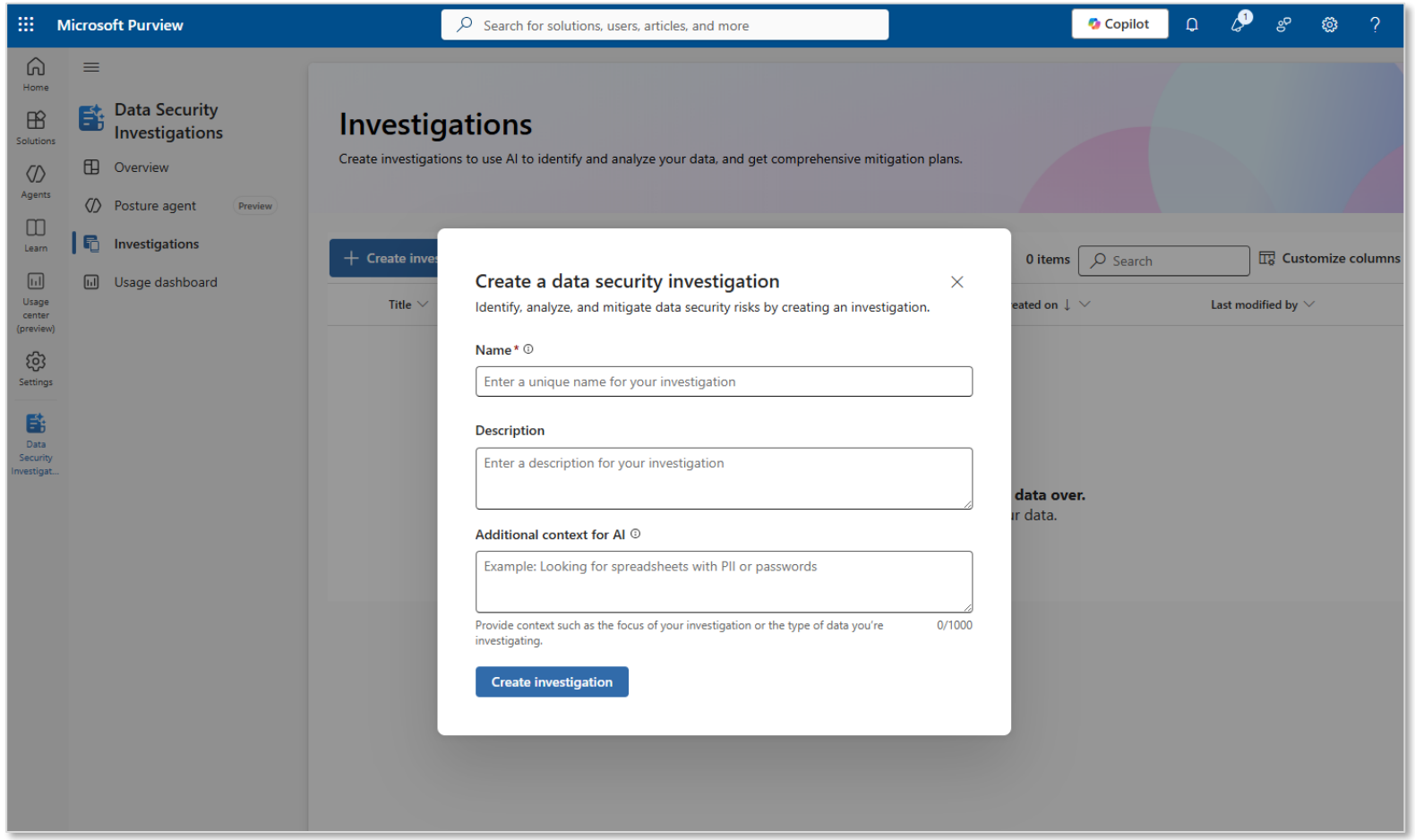
Introducing Purview Data Security Investigations (DSI)



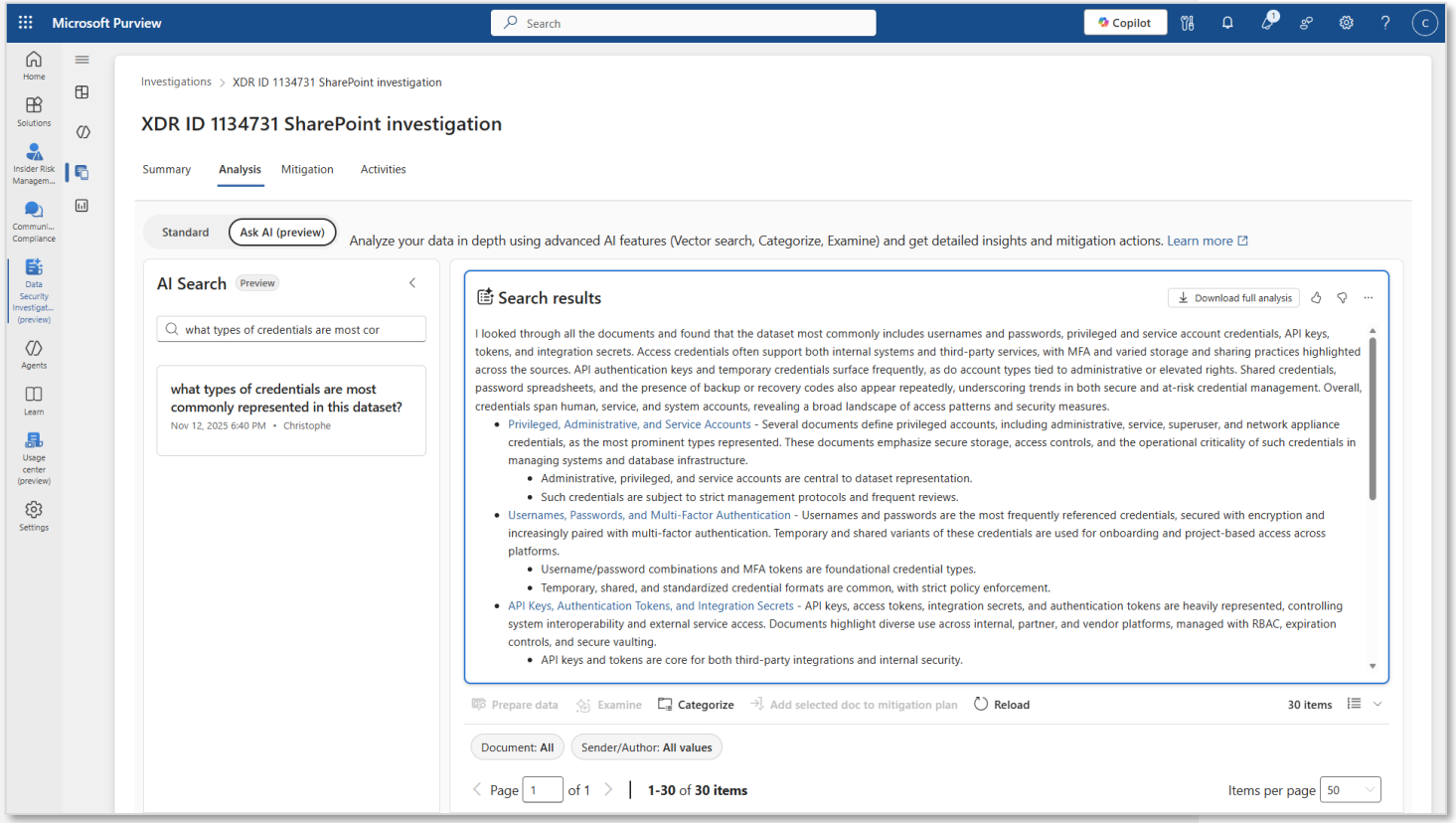
- Focuses investigations on content, not just activity
- Leverages generative AI for deep data understanding
- Designed for both reactive and proactive scenarios

What Makes Purview DSI different?

- AI-driven search
- Deep content analysis
- Data categorization
- Summarization of impacted data
- Audit log search
- Built-in mitigations



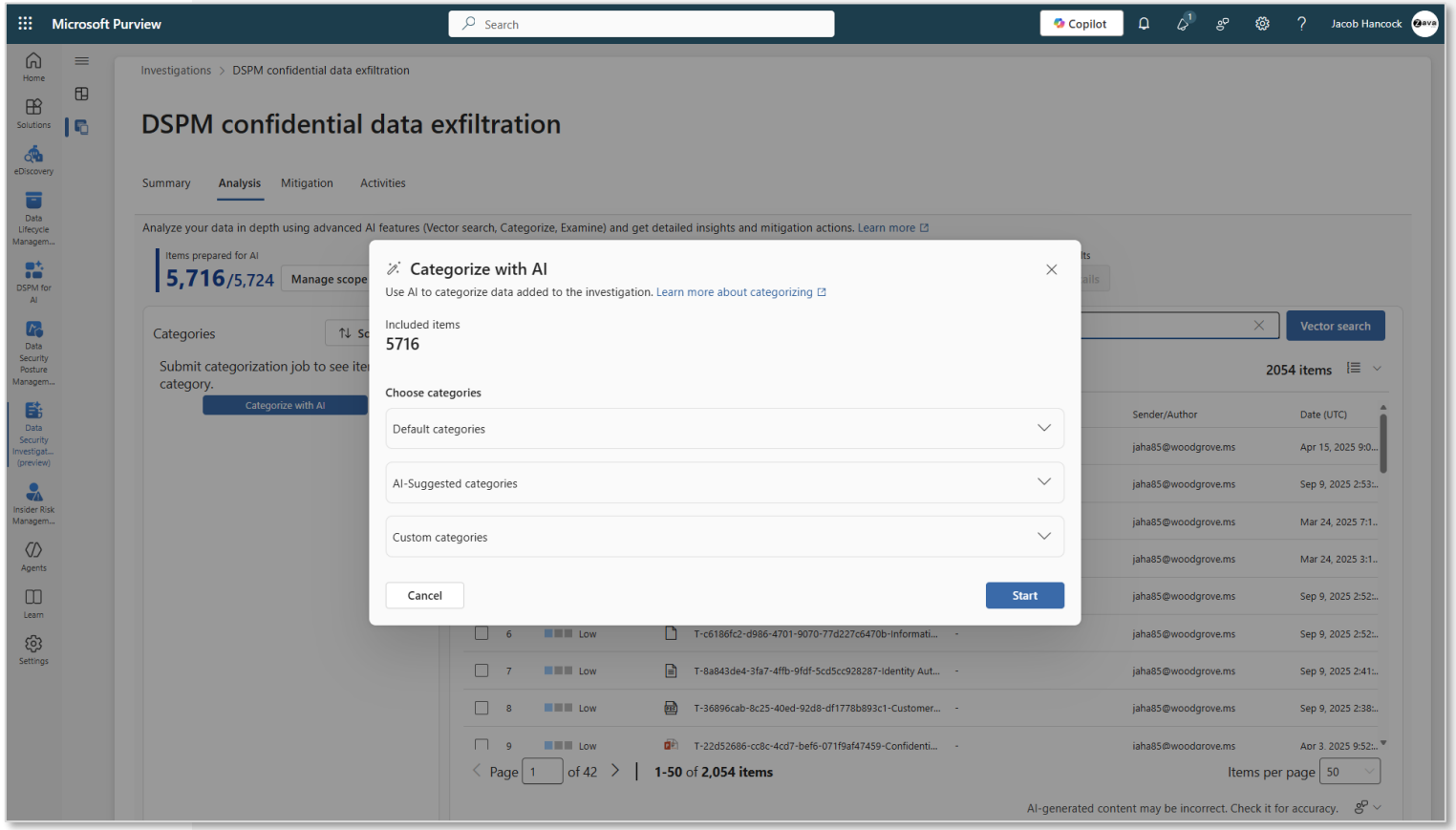
AI Search



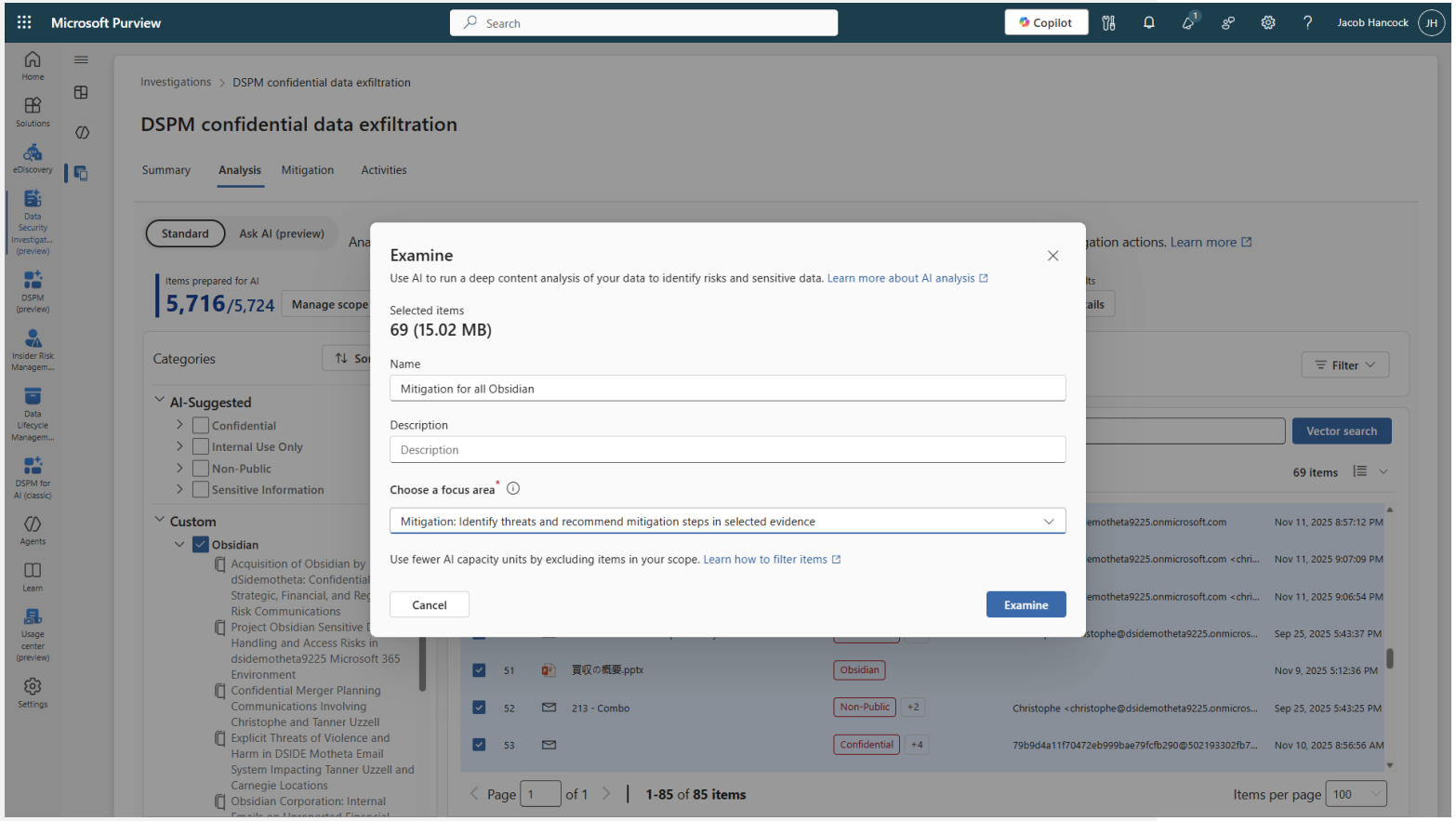
- Reason over data using natural language
- Discover relevant content even when keywords are missing
- Faster identification of investigation-relevant data

Categorization

- Automatically categorizes impacted data
- Groups content by subject and risk
- Helps prioritize high-risk data quickly



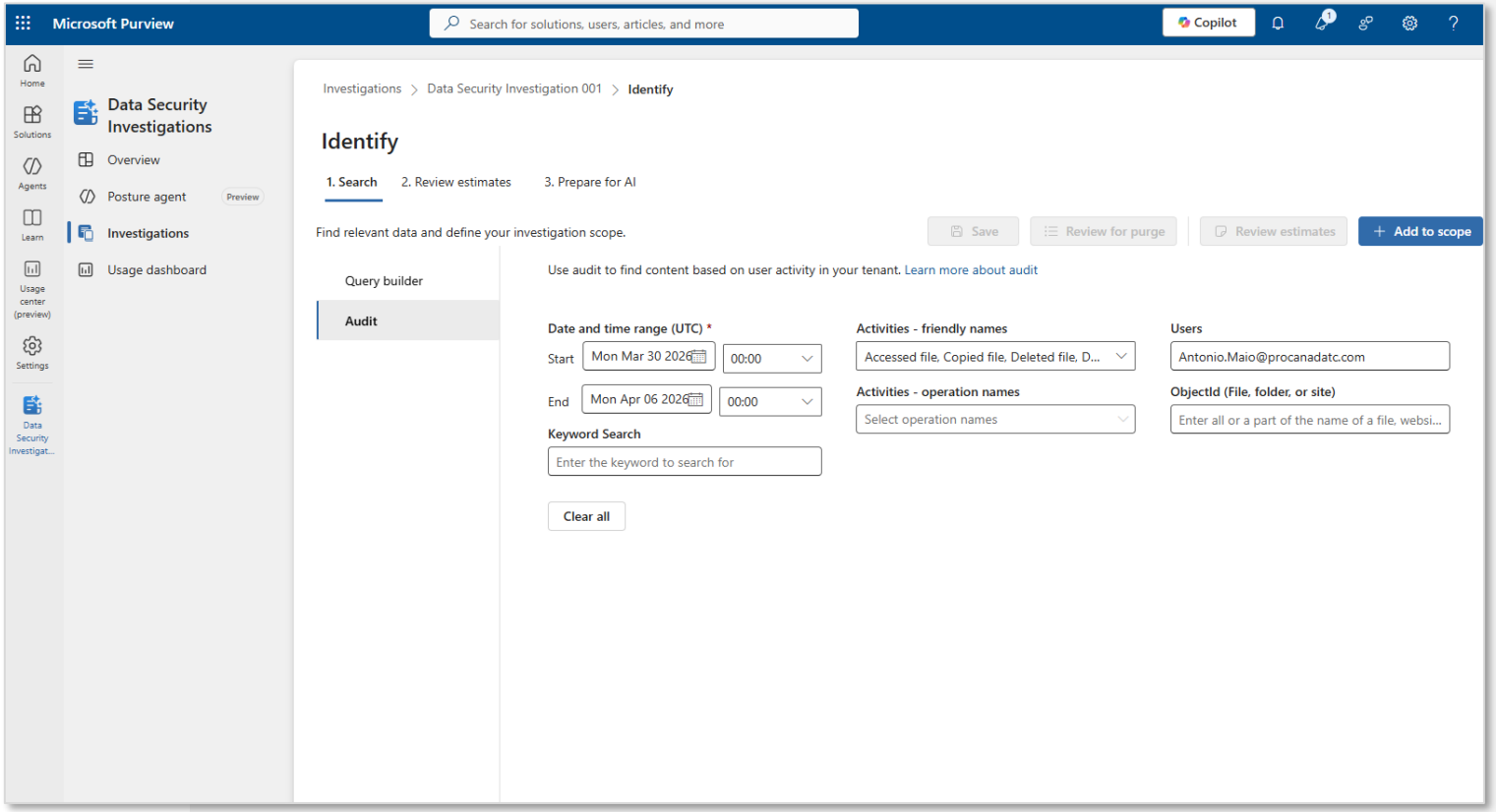
Examination



- Examine content for security risks
- Identify sensitive information and exposures
- Surface insights that are hard to detect manually

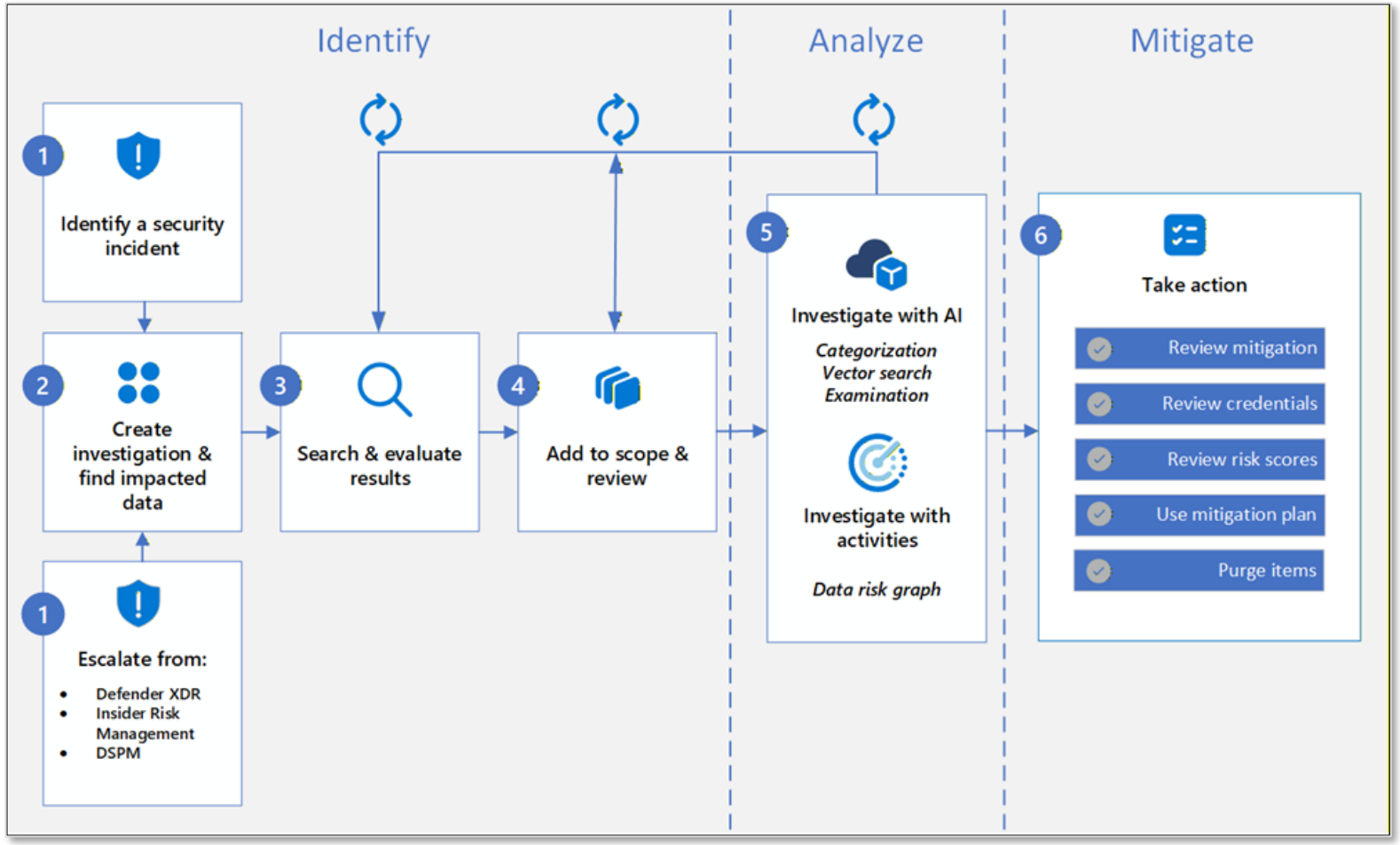
Audit Log Search

- Incorporate Audit Log activities & search results into a DSI investigation
- Corroborate AI Search result data with user activities



Why Generative AI Changes Everything

- Analyze large volumes of data quickly
- Reduce manual review effort
- Enable informed decisions earlier in an investigation



Purview DSI Roles & Use Cases

Data Security Investigation (DSI) Roles

Purview Role	Description of Capabilities
DSI Admins	Can administer and manage all features related to DSI, including configuring Azure Pay-as-you-Go licensing requirements.
DSI Investigators	Can **create** and **manage** investigations to which they are assigned can do all other activities within those investigations; Note: This role cannot configure Azure Pay-as-you-Go licensing requirements.
DSI Reviewers	Can view investigations to which they are assigned, but this is not simply a read-only view; • Can also add, delete, and manage migration plan items, • Can run categorization activities, examination activities; vector searches and view data risk graphs. Note: These capabilities will affect your Azure subscription consumption.

Proactive Use Cases

DSI can help:

- Discover high-risk data exposures or conduct
- Identify sensitive content in unexpected locations
- Conducting assessments or risk Audits to reduce risk through early remediation
- Overcome “needle in a haystack” challenges

Proactive Use Case 1: Discover Inappropriate Conduct

- **Scenario**: As part of a periodic assessment or compliance audit, we are conducting searches to validate whether inappropriate conduct is happening.
  - This scenario is traditionally thought of as reactive but can also be proactive, particularly if using keywords or identifiers within a large data estate to discovery activity that could potentially become concerning / malicious or contrary to policies and regulations.
- DSI's AI search allows us ask natural language questions or enter keywords with a specific focus to narrow down items for review.
- DSI's examination feature enables deep content analysis on selected data looking for security risks (e.g.: credentials, or evidence of discussion).
  - Once security risks are identified, you can scan for sensitive data (e.g.: personal data, financial, or health information).

Proactive Use Case 2: Assess or Audit

- **Scenario**: With increasing use of GenAI tools like Microsoft 365 Copilot, organizations face new risks related to the prompts users submit, and the AI-generated responses they receive.
  - These interactions may inadvertently include or expose sensitive data such as customer information, intellectual property, or regulated content.

- DSI enables SecOps teams to proactively audit AI prompt and response logs and use AI-powered examination and categorization to detect:
  - Prompts containing sensitive internal data (e.g., customer records, financial data, source code).
  - Responses that may include hallucinated or misrepresented sensitive information.
  - Patterns of risky behavior (e.g., repeated attempts to extract confidential data via AI).

Reactive Use Cases

DSI can help:

- Rapidly assess impacted data
- Understand true business impact
- Focus response and mitigation efforts

Reactive Use Case 1: Investigate a Data Breach

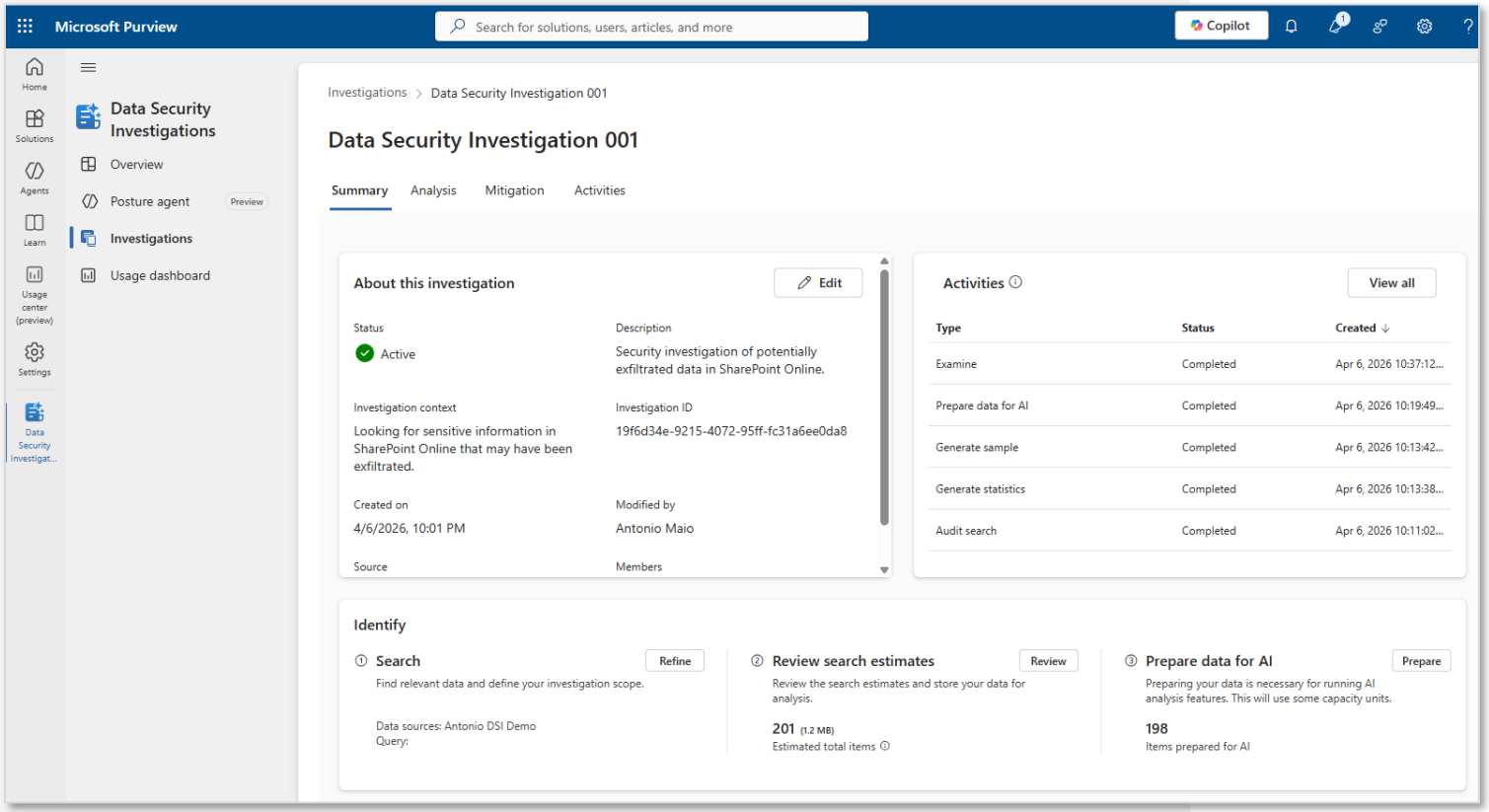
- **Scenario**: a data breach has already occurred, and we're tasked with investigating impact, determining root cause, and determining containment actions.
- DSI enables SecOps teams to understand the nature, scope, impact of the data that was potentially included in the breach.
  - In a post-breach scenario, "time is money". DSI features like AI search, categorization, and examination help us to understand the nature of that data at a much deeper level and more quickly than we ever could.
  - Categorization helps us understand the general trends of the data that was exposed.
  - AI search helps us understand more precisely what type of sensitive data may have been included in exposed documents.
- This allows us to focus investigations on the most important data and report on organizational risks much more precisely to leadership.

Reactive Use Case 2: Disrupted Attack Root Cause Analysis

- **Scenario**: an attack is stopped by *Microsoft Defender XDR automatic attack disruption*, and SecOps analysts need to investigate the root cause of the attack to ensure that the attack path is well understood (by leveraging *Microsoft Security Exposure Management*) and effective mitigation steps are taken to prevent it occurring in the future.
  - We need to understand the nature of target data
  - We need to ensure target data is stored in the correct repositories (or recommend moving if not) and validate that effective security controls are enforced on the target data.
- A DSI can be started from an incident (escalated) within Microsoft Defender XDR, enabling investigators to bring in mailbox, email message, or file node data.
  - The items included in the scope of the incident are automatically included as data sources. You can start reviewing and adding these to the investigation scope in DSI.

Key Takeaways

The Purview DSI Investigation Experience



- Centralized investigation experience
- Securely collaborate on sensitive findings
- Avoid secondary incidents from oversharing

Why Purview DSI Matters & Looking Ahead…

- Shifts investigations from logs to focus on Data
- Enables proactive and reactive security scenarios
- Uses AI to bring clarity, speed & scale to investigations
- Creates a foundation for deeper AI-driven investigations
- Check out our blog for a first look at Purview DSI in more detail: https://bit.ly/4cfYIw0

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